CLOSED SUBSETS OF A NOETHERIAN SPACE CHARACTERISED BY STABILITY UNDER GENERALIZATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of constructible subsets of a Noetherian space that contain an open of every irreducible closed subset meeting them. Invoked inside the counted proof (source lines 4030-4032). Shows the subset contains an open of each irreducible closed subset meeting it.
- The characterization of open subsets of a Noetherian space through that property. Invoked inside the counted proof (source lines 4033-4034). Turns that property into openness.

Source reference. Original lemma lemma-characterize-constructible-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 3113–3113. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-constructible-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-characterize-open-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 3200–3200. Navigation pointer only: the counted text refers to the Stacks item lemma-characterize-open-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a Noetherian sober topological space. Let $E \subset X$ be a subset of X.*

(1) *If E is constructible and stable under specialization, then E is closed.*
(2) *If E is constructible and stable under generalization, then E is open.*

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Proof. Let E be constructible and stable under generalization. Let $Y \subset X$ be an irreducible closed subset with generic point $\xi \in Y$. If $E \cap Y$ is nonempty, then it contains ξ (by stability under generalization) and hence is dense in Y, hence it contains a nonempty open of Y, see Lemma lemma-characterize-constructible-Noetherian. Thus E is open by Lemma lemma-characterize-open-Noetherian. This proves (2). To prove (1) apply (2) to the complement of E in X. $\square$